Jinuk Moon

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Seoul National University

Profile

PhD candidate developing machine learning interatomic potentials (MLIPs) for accelerating catalyst discovery. Developed CatBench, a benchmarking framework for reliable evaluation of MLIPs in heterogeneous catalysis, and oh-my-mlip, an LLM agent-driven framework for MLIP installation, benchmarking, fine-tuning, and distillation. Research focuses on knowledge distillation of foundation MLIPs and multiscale simulation combining machine learning with first-principles calculations to discover efficient catalytic materials for clean energy applications.

Education

- **Seoul National University** *2023 – Present*
  Ph.D. Materials Science and Engineering Seoul, South Korea
  - Machine learning-accelerated computational catalysis for electrocatalyst discovery and design
  - GPA: 4.15/4.3

- **Pohang University of Science and Technology (POSTECH)** *2019 – 2022*
  B.S. Chemical Engineering Pohang, South Korea
  - Thesis: Study of Metal-Nitrogen-doped carbon catalysts for Chlorine Evolution Reaction combining DFT calculation with Machine Learning
  - GPA: 3.98/4.3 (Summa Cum Laude)

Open Source & Code

- **CatBench - Benchmark Framework for Machine Learning Interatomic Potentials in Catalysis**
  Standardized framework for fair comparison of MLIPs in heterogeneous catalysis
  - pip-installable Python package | catbench.org | github.com/JinukMoon/CatBench

- **oh-my-mlip - Unified Hub for Machine Learning Interatomic Potentials**
  LLM agent-driven framework for MLIP installation, execution, benchmarking, fine-tuning, and distillation
  - Agent plugin | oh-my-mlip.org | github.com/JinukMoon/oh-my-mlip

Honors & Awards

- **Presidential Science Scholarship (PhD Track; KRW 24M/yr, ∼USD 17K/yr)** *2025 – Present*
  Ministry of Science and ICT South Korea
- **Soseon Foundation Scholarship (KRW 20M, ∼USD 14K)** *2026*
  Soseon Foundation – Graduate research scholarship for outstanding STEM students South Korea
- **Outstanding Research Achievement Award** *Fall 2026*
  BK21 Creative Materials Education and Research Group, Seoul National University South Korea
- **Best Oral Award** *Spring 2025*
  The Korean Institute of Metals and Materials (KIM) South Korea
  - CatBench: Benchmark of Machine Learning Interatomic Potentials for Adsorption Energy Predictions in Heterogeneous Catalysis

- **Best Oral Award** *Fall 2024*
  The Korean Institute of Metals and Materials (KIM) South Korea
  ◦ Finding Optimal MoC Catalyst for Hydrogen Evolution via Fine-Tuned Machine Learning Potential

- **Best Oral Award** *Spring 2024*
  The Korean Ceramic Society (KCerS) South Korea
  ◦ Selective Control of Oxygen Evolution Mechanisms by Doping Concentrations in RuO_2

- **Best Poster Award** *2025*
  The 8th International Conference on Advanced Electromaterials (ICAE 2025) South Korea
  ◦ Machine Learning-Accelerated Multiscale Modeling of Metal/MoC Electrocatalysts for Alkaline Hydrogen Evolution

PUBLICATIONS (13 PUBLISHED, 2 SUBMITTED, 9 FIRST-AUTHORED)

† Equal contribution, * Corresponding author

1st **[submitted]** **Moon, J.**†, Kim, H.†, Kim, T. Y., Bae, J.*, Han, J. W.* (2026). Large language models across the catalyst development workflow: applications in experiment and computation. *Nano-Micro Letters* (submitted).

1st **[submitted]** Jo, H.†, Sohn, Y.†, **Moon, J.**†, Han, J. W.*, Kim, W.*, Yong, K.* (2026). Synthesis-Stage Interfacial Engineering of Edge-Rich β-$Co(OH)_2$ for Efficient Nitrate to Ammonia Electrosynthesis. *ACS Nano* (under review).

1st **[1]** Jun, H.†, Kwon, D.†, **Moon, J.**†, Han, S.†, Kim, S., Yi, S. Y., Kim, Y., Kang, E., Choi, C. H.*, Han, J. W.*, Kim, W.*, Lee, J.* (2026). Synergistic Effect of Tuning Reaction Mechanism and Interfacial Water for Efficient Acidic Water Oxidation. *Small*, in press.

1st **[2]** **Moon, J.**, Han, J. W.* (2026). Charge-aware knowledge distillation of foundation interatomic potentials. *AI for Accelerated Materials Design (AI4Mat) Workshop, NeurIPS 2026*.

1st **[3]** Kim, J.†, **Moon, J.**†, Cho, K.†, Kim, J.†, Han, M. H., Kim, S., Han, J., Choi, D., Jun, H., Jang, J. H., Oh, H.-S., Cho, S. K.*, Kim, W.*, Han, J. W.*, Lee, J.* (2026). Size-Dependent Dynamic Surface States of Ru Clusters: Modulating Interfacial Water Structure for Enhanced Alkaline Hydrogen Evolution Reaction. *Energy & Environmental Science*, 19, 4211.

[4] Kim, J.†, Yang, G. G.†, Lee, W.†, Nam, S., **Moon, J.**, Yang, H., Seo, J., Choi, Y., Jeong, B.*, Han, J. W.*, Kim, S. O.*, Jung, W.* (2026). Quantitative Electrochemical Evaluation of Metal Nanocatalysts for Oxygen Exchange on Solid Oxide Cell Electrodes. *Energy & Environmental Science*, 19, 4932.

[5] Lim, H.†, Choung, S.†, **Moon, J.**, Han, J. W.* (2026). Angular relational knowledge distillation of machine learning interatomic potentials for scalable catalyst exploration. *npj Computational Materials*, 12, 193.

[6] Park, K.†, Lee, W.†, Park, J., Jo, M., Bae, Y., Kim, M., Shin, S., **Moon, J.**, O'Hayre, R.*, Han, J. W.*, Song, S.*, Park, J. Y.* (2026). Composite air electrodes based on $BaCe_{0.7}Zr_{0.1}Y_{0.1}Yb_{0.1}O_{3-\delta}$ for reversible solid oxide cells. *Nature Energy*, 11, 996.

[7] Jung, H.†, Lee, Y.†, Cho, K.†, Kim, T. Y., Song, J., Jung, H. S., Park, S. I., Lee, Y., **Moon, J.**, Park, W., Nam, J., Park, S., Kim, W., Han, J. W.* (2026). Strategic Design of Oxophilic Dopants for Active and Durable Alkaline Hydrogen Evolution Reaction Under Seawater. *Advanced Functional Materials*, 36, e16387.

1st **[8]** Choung, S.†, Kim, M.†, **Moon, J.**†, Han, J. W.* (2025). From Atomic Motif to Realistic Single Atom Catalysts through Machine Learning Interatomic Potentials. *ACS Energy Letters*, 10, 6288.

1st **[9]** **Moon, J.**, Jeon, U., Choung, S., Han, J. W.* (2025). CatBench Framework for Benchmarking Machine Learning Interatomic Potentials in Adsorption Energy Predictions for Heterogeneous Catalysis. *Cell Reports Physical Science*, 6, 102968.

1st **[10]** Jun, H.†, Kang, E.†, **Moon, J.†**, Kim, H., Han, S., Choung, S., Kim, S., Yi, S. Y., Kang, E., Choi, C. H.*, Han, J. W.*, Lee, J.* (2025). Quantity effect of heteroatom incorporation on the oxygen evolution mechanism in ruthenium oxide. *Chem*, 11(5), 102367.

1st **[11]** Choung, S.†, Park, W.†, **Moon, J.†**, Han, J. W.* (2024). Rise of machine learning potentials in heterogeneous catalysis: Developments, applications, and prospects. *Chemical Engineering Journal*, 494, 152757.

[12] Choung, S.†, Yang, H.†, **Moon, J.**, Park, W., June, H., Lim, C., Han, J. W.* (2024). Theoretical tuning of local coordination environment of metal-nitrogen doped carbon catalysts for selective chlorine-evolution reaction. *Catalysis Today*, 425, 114358.

[13] Choung, S., Kim, Y., **Moon, J.**, Roh, J., Hwang, J., Han, J. W.* (2023). Unveiling the catalyst deactivation mechanism in the non-oxidative dehydrogenation of light alkanes on Rh(111): Density functional theory and kinetic Monte Carlo study. *Catalysis Today*, 411-412, 113819.

Conference Presentations & Seminars

[1] Invited **Moon, J.** (2027). Machine Learning Interatomic Potentials on HPC: From Fine-tuning to Distillation (Webinar). *ENCCS & Sweden AI Factory (EuroHPC)*, Online, Jan. 21, 2027.

[2] Invited **Moon, J.** (2026). Machine Learning Interatomic Potentials on HPC: From Installation to Benchmarking (Webinar). *ENCCS & Sweden AI Factory (EuroHPC)*, Online, Dec. 1, 2026.

[3] **Moon, J.** (2026). Charge-aware Knowledge Distillation of Machine Learning Interatomic Potentials (Oral Presentation). *Korean Institute of Metals and Materials (KIM)*, Daejeon, South Korea.

[4] **Moon, J.** (2026). Charge-aware Knowledge Distillation of Machine Learning Interatomic Potentials (Poster). *AI4Sci Korea 2026*, Seoul, South Korea.

[5] **Moon, J.** (2026). CatBench: Evaluating Accuracy, Robustness, and Efficiency of Machine Learning Interatomic Potentials for Adsorption Energy Predictions (Oral Presentation). *American Chemical Society (ACS)*, Chicago, United States.

[6] **Moon, J.** (2026). Size-Dependent Surface States of Sub-Nanometer Ru Clusters Modulate Interfacial Water Structure for Enhanced Alkaline Hydrogen Evolution (Poster). *American Chemical Society (ACS)*, Chicago, United States.

[7] **Moon, J.** (2026). Size-Dependent Surface States of Sub-Nanometer Ru Clusters Modulate Interfacial Water Structure for Enhanced Alkaline Hydrogen Evolution (Poster, Sci-Mix). *American Chemical Society (ACS)*, Chicago, United States.

[8] **Moon, J.** (2026). Synergistic Effect of Oxygen Evolution Mechanism and Interfacial Water Dynamics Revealed by MLIP Distillation (Oral Presentation). *Korean Institute of Metals and Materials (KIM)*, Jeju, South Korea.

[9] **Moon, J.** (2026). Machine Learning-Accelerated Large-Scale Analysis of Interfacial Water Structure on Size-Controlled Ru Clusters for Alkaline Hydrogen Evolution (Poster). *The Korean Society of Industrial and Engineering Chemistry (KSIEC)*, Jeju, South Korea.

[10] **Moon, J.** (2025). Machine Learning-Accelerated Multiscale Modeling of Metal/MoC Electrocatalysts for Alkaline Hydrogen Evolution (Poster). *The 8th International Conference on Advanced Electromaterials (ICAE 2025)*, Jeju, South Korea.

[11] **Moon, J.** (2025). Machine Learning-Accelerated Multiscale Modeling of Metal/MoC Electrocatalysts for Alkaline Hydrogen Evolution (Poster). *Korean Institute of Metals and Materials (KIM)*, Gwangju, South Korea.

[12] **Moon, J.** (2025). CatBench: Benchmark Framework of Machine Learning Interatomic Potentials for Adsorption Energy Predictions in Heterogeneous Catalysis (Poster). *The 29th North American Catalysis Society Meeting (NAM29)*, Atlanta, United States.

[13] **Moon, J.** (2025). Multiscale Modeling of Nanostructured Electrocatalyst for Alkaline Water Electrolysis via Graph Neural Network (Oral Presentation). *The 29th North American Catalysis Society Meeting (NAM29)*, Atlanta, United States.

[14] **Moon, J.** (2025). Multiscale Modeling of Nanostructured Electrocatalyst for Alkaline Water Electrolysis via Fine-Tuned Machine Learning Interatomic Potential (Oral Presentation). *Korean Institute of Chemical Engineers (KICHE)*, Jeju, South Korea.

[15] **Moon, J.** (2025). CatBench: Benchmark Framework of Machine Learning Interatomic Potentials for Adsorption Energy Predictions in Heterogeneous Catalysis (Poster). *The Korean Society of Industrial and Engineering Chemistry (KSIEC)*, Jeju, South Korea.

[16] **Moon, J.** (2025). CatBench: Benchmark of Machine Learning Interatomic Potentials for Adsorption Energy Predictions in Heterogeneous Catalysis (Oral Presentation). *Korean Institute of Metals and Materials (KIM)*, Jeju, South Korea.

[17] **Moon, J.** (2024). Finding Optimal MoC Catalyst for Hydrogen Evolution via Fine-Tuned Machine Learning Potential (Oral Presentation). *Korean Institute of Metals and Materials (KIM)*, PyeongChang, South Korea.

[18] **Moon, J.** (2024). Finding Optimal MoC Catalyst for Hydrogen Evolution via Fine-Tuned Machine Learning Potential (Oral Presentation). *23rd Korea Japan Students' Symposium*, Seoul, South Korea.

[19] **Moon, J.** (2024). Selective Control of Oxygen Evolution Reaction Mechanisms by Adjusting Mn Doping Concentrations in RuO_2 (Oral Presentation). *The Korean Ceramic Society (KCerS)*, Busan, South Korea.

Skills

- **Programming & Frameworks:** Python (Advanced), PyTorch, Bash scripting (Intermediate)
- **Machine Learning:** MLIP Training & Benchmarking (20+ architectures), Dataset Generation, Bayesian Optimization
  - *MLIP Fine-tuning:* Generated 100,000+ in-house DFT calculations, fine-tuned GemNet-OC reducing adsorption energy MAE from 0.48 eV to 0.05 eV (evaluated by CatBench)
  - *Large-scale Molecular Dynamics:* MLIP-accelerated MD simulations (100+ ns) for metal dispersion and structural evolution analysis
  - *High-throughput Screening:* MLIP-accelerated screening of alloy catalysts using bulk structures from Materials Project, OQMD, GNoMe, and OMat datasets
  - *Genetic Algorithm:* Ensemble-GA development with sub-population partitioning and mixing strategies for enhanced global optimization
- **Computational Chemistry:** Density Functional Theory (VASP), Machine Learning Interatomic Potentials (20+ models: UMA, EquiformerV2, GemNet-OC, MACE, etc.), Molecular Dynamics, Electronic Structure Analysis
- **High-Performance Computing:** Supercomputer Cluster Server Management, SLURM, PBS, Batch Job Optimization
- **Research Expertise:** Heterogeneous Catalysis, Single Atom Catalysis, Electrocatalysis (OER, HER, CER), Multiscale Simulation using MLIP, MLIP Benchmarking, ML Analysis of Experimental Characterization
- **Language:** Korean (Native), English (Fluent)

Teaching

- **Seoul National University** — *Fall 2025, Fall 2026*
  TA - Computer Concepts and Practice — Seoul, South Korea
- **Seoul National University** — *Fall 2024*
  TA - Computational Materials and Data Science — Seoul, South Korea
- **Seoul National University** — *Fall 2023*
  TA - Physical Chemistry of Materials — Seoul, South Korea